CSE 400: Joint Random Variables

Lecture 16 Notes

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Topic 1: Introduction to Joint Random Variables

1. Why Study Joint Random Variables?

In real-life and scientific applications, we are often interested in two or more random variables (RVs) simultaneously. Relationships are expected between simultaneous variables, such as vehicle speed and traffic density.

2. Definitions and Notation

- **Joint Distribution:** A distribution that allows for the computation of probabilities for events involving two or more variables simultaneously and provides an understanding of their relationship.
- **Independence:** The simplest state of joint variables where they do not influence each other.
- **Covariance:** A measure of the nature of dependence between non-independent variables.
- **Correlation:** A measure used to understand the relationship between non-independent variables.

3. Main Results and Application Areas

Correlation Example: Variables X (Signal-to-Noise Ratio) and Y (Data Throughput) are correlated because throughput depends on SNR.

Applications:

- **Smart Transportation:** Adaptive signals, congestion prediction, and autonomous routing.
- **Wireless Networks:** 5G scheduling, adaptive modulation, and network planning.
- **Other:** Weather forecasting, UAV logistics, and social science.

4. Worked Examples

Example 2: Wireless Network Performance
X = SNR Level, Y = Data Throughput

SNR Level (X)	Throughput High	Throughput Low
High	0.40	0.10
Medium	0.20	0.15
Low	0.05	0.10

Topic 2: Discrete Joint Random Variables

1. Joint Probability Mass Function (PMF)

Setup: $X \in \{x_1, \dots, x_n\}$ and $Y \in \{y_1, \dots, y_m\}$.

- **Product Set:** The pair (X, Y) takes values in $\{(x_1, y_1), \dots, (x_n, y_m)\}$.
- **Joint PMF:** $p(x_i, y_j) = P(X = x_i, Y = y_j)$.

2. Conditions for Validity

1. **Non-negativity:** $0 \le p(x_i, y_j) \le 1$.
2. **Normalization:** $\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$.

3. Worked Example: Rolling Two Dice

X = value on first die, T = total sum.

- $P(X = 1, T = 2) = 1/36$
- $P(X = 1, T = 8) = 0$
- $P(X = 2, T = 3) = 1/36$

Observation: Entries are either 0 or 1/36. X and T are dependent.

Topic 3: Continuous Joint Random Variables

1. Joint Probability Density Function (PDF)

Definition: A function $f(x, y)$ such that $P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$.

- **Non-negativity:** $f(x, y) \ge 0$.
- **Normalization:** $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$.

2. Worked Example

Given $f(x, y) = 4xy$ for $0 \le x, y \le 1$.

Normalization Check:

$$\int_0^1 \int_0^1 4xy\, dx\, dy = \int_0^1 [2x^2 y]_0^1\, dy = \int_0^1 2y\, dy = [y^2]_0^1 = 1$$

Compute $P(X < 0.5, Y > 0.5)$**:**

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy\,dy\,dx = \int_0^{0.5} [2xy^2]_{0.5}^{1}\,dx = \int_0^{0.5} 1.5x\,dx = [\frac{3}{4}x^2]_0^{0.5} = \frac{3}{16}$$

Topic 4: Joint Cumulative Distribution Function (CDF)

1. Definition

$F(x, y) = P(X \leq x, Y \leq y)$.

- **Continuous:** $F(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v)\,du\,dv$.
- **Discrete:** $F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$.

2. Properties

- **Recovering PDF:** $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$.
- **Limits:** $F(-\infty, -\infty) = 0$ and $F(\infty, \infty) = 1$.
- **Monotonicity:** $F(x, y)$ is non-decreasing in both x and y.